

• Medium • #6 • ZigZag Conversion

▲ **Problem Statement** • Given a string "a" & an integer "numRows", arrange the characters in a zigzag pattern & then read the characters row by row.

Ex: P A H N
 A P L S I I G ⇒ PAYPAL IS HIRING
 Y I R

▲ **Main Problem Solution** • Simulate the zigzag pattern using a vector of strings.

Each character is added to the current ~~row~~ row.

- Move down through the rows
- When we reach the bottom, change direction & move up.
- When we reach the top, change direction again.

Finally, concatenate all rows.

▲ Brute Force Method • Actually construct the entire zigzag pattern including the empty spaces, then read it row by row.

This uses unnecessary space for the empty positions.

Time: $O(n)$

Space: $O(n \times \text{numRows})$

▲ Algorithm → 1 | If " $\text{numRows} == 1$ " or " $\text{numRows} \geq s.\text{length}$ ", return " s ".

2 | Create " numRows " strings to ~~present~~ represent each row.

3 | Start at row " 0 " & move backward.

4 | Add each character to the current row.

5 | When reaching the top or bottom, reverse the direction.

6 | Continue until all characters are processed.

7 | Concatenate all rows.

8 | Return the result.

▲ Important Points • Use " $\text{vector} < \text{string} >$ " to store characters belonging to each row.

• " $\text{direction} = 1$ " → moving down

- "direction = -1" $\rightarrow$ moving up
- Change direction at the top & bottom rows.
- No need to actually create the spaces in the zigzag pattern.
- "numRows == 1" is an important edge case.

▲ Complexity: Time: $O(n)$
Space: $O(n)$